Introduction to Algorithms

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Designing Algorithms

- Insertion sort uses the incremental method: for each element A[i], insert it into its proper place in the subarray A[1:i], having already sorted the subarray A[1:i-1].
- This section examines another design method, known as divide-and-conquer.
- Worst-case running time for divide-and-conquer is much less than that of insertion sort.
- Analyzing its running time is often straightforward.

The divide-and-conquer method

- Many useful algorithms are recursive in structure: to solve a given problem, they recurse (call themselves) one or more times to handle closely related subproblems.
- They break the problem into several subproblems that are similar to the original problem but smaller in size, solve the subproblems recursively, and then combine these solutions to create a solution to the original problem.
- Base case: if the problem is small enough-the base Case- you just solve it directly without recursing.

The divide-and-conquer method

- Recursive case:

- **Divide** the problem into one or more subproblems that are smaller instances of the same problem.

- **Conquer** the subproblems by solving them recursively.

- **Combine** the subproblem solutions to form a solution to the original problem.

The merge sort algorithm

- **Divide** the subarray A[p:r] to be sorted into two adjacent subarrays, each of half the size. To do so, compute the midpoint q of A[p:r] (taking the average of p and r), and divide A[p:r] into subarrays A[p:q] and A[q+1:r].

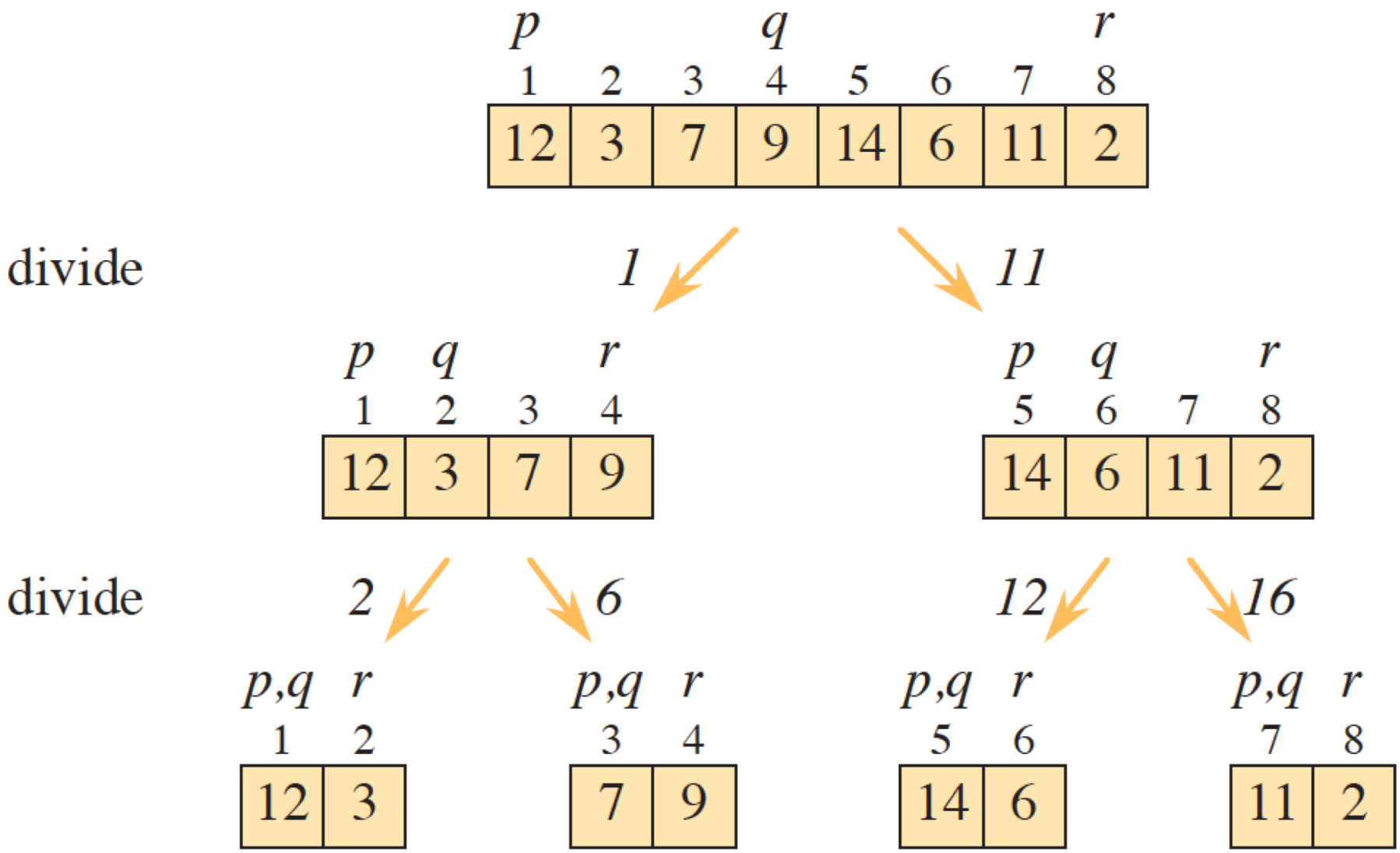
- **Conquer** by sorting each of the two subarrays A[p:q] and A[q+1:r] recursively using merge sort.

- **Combine** by merging the two sorted subarrays A[p:q] and A[q+1:r] back into A[p:r], producing the sorted answer.

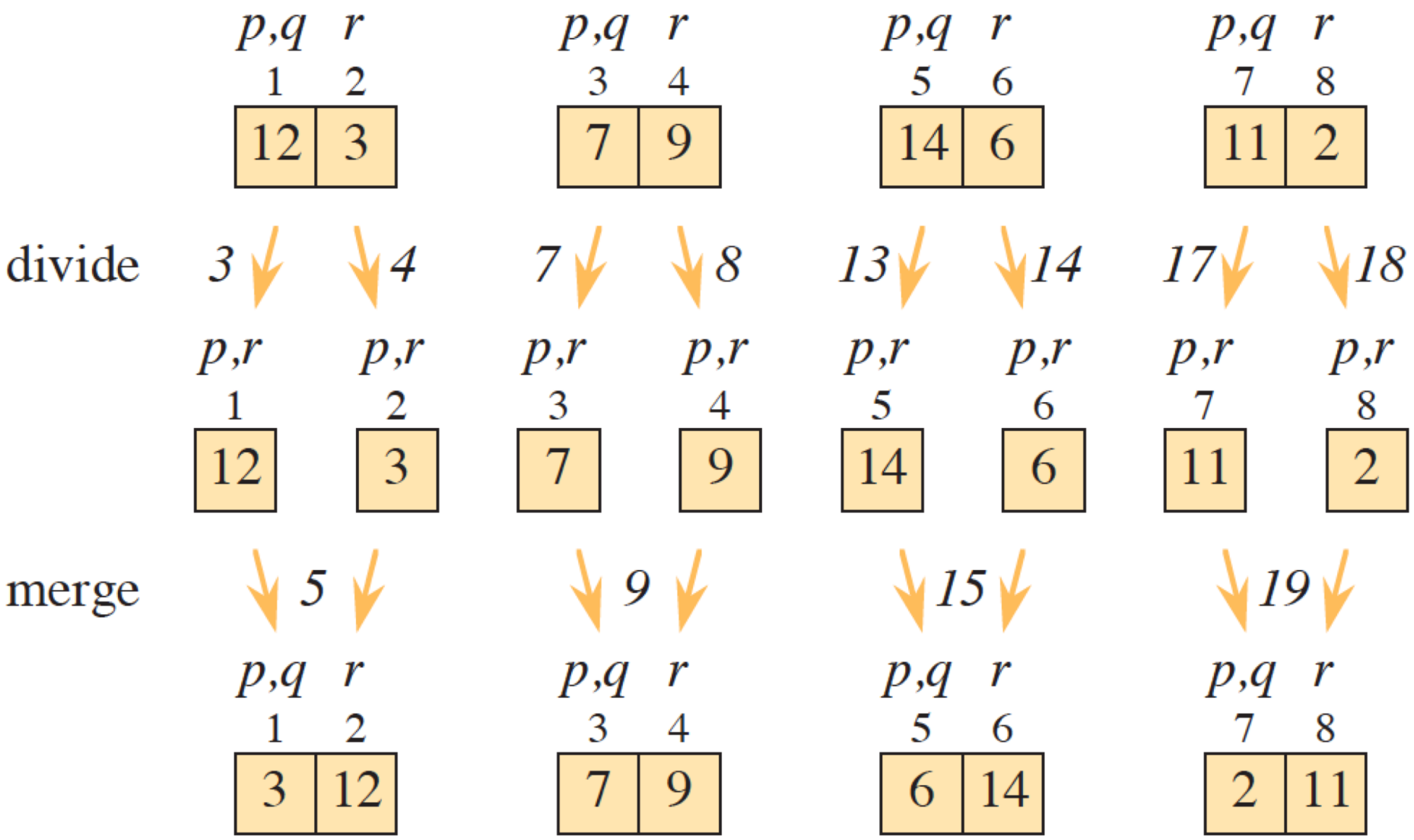
The procedure MERGE-SORT

```
MERGE-SORT(A, p, r)
1  if p ≥ r                                  // zero or one element?
2      return
3  q = ⌊(p + r)/2⌋                           // midpoint of A[p : r]
4  MERGE-SORT(A, p, q)                       // recursively sort A[p : q]
5  MERGE-SORT(A, q + 1, r)                   // recursively sort A[q + 1 : r]
6  // Merge A[p : q] and A[q + 1 : r] into A[p : r].
7  MERGE(A, p, q, r)
```

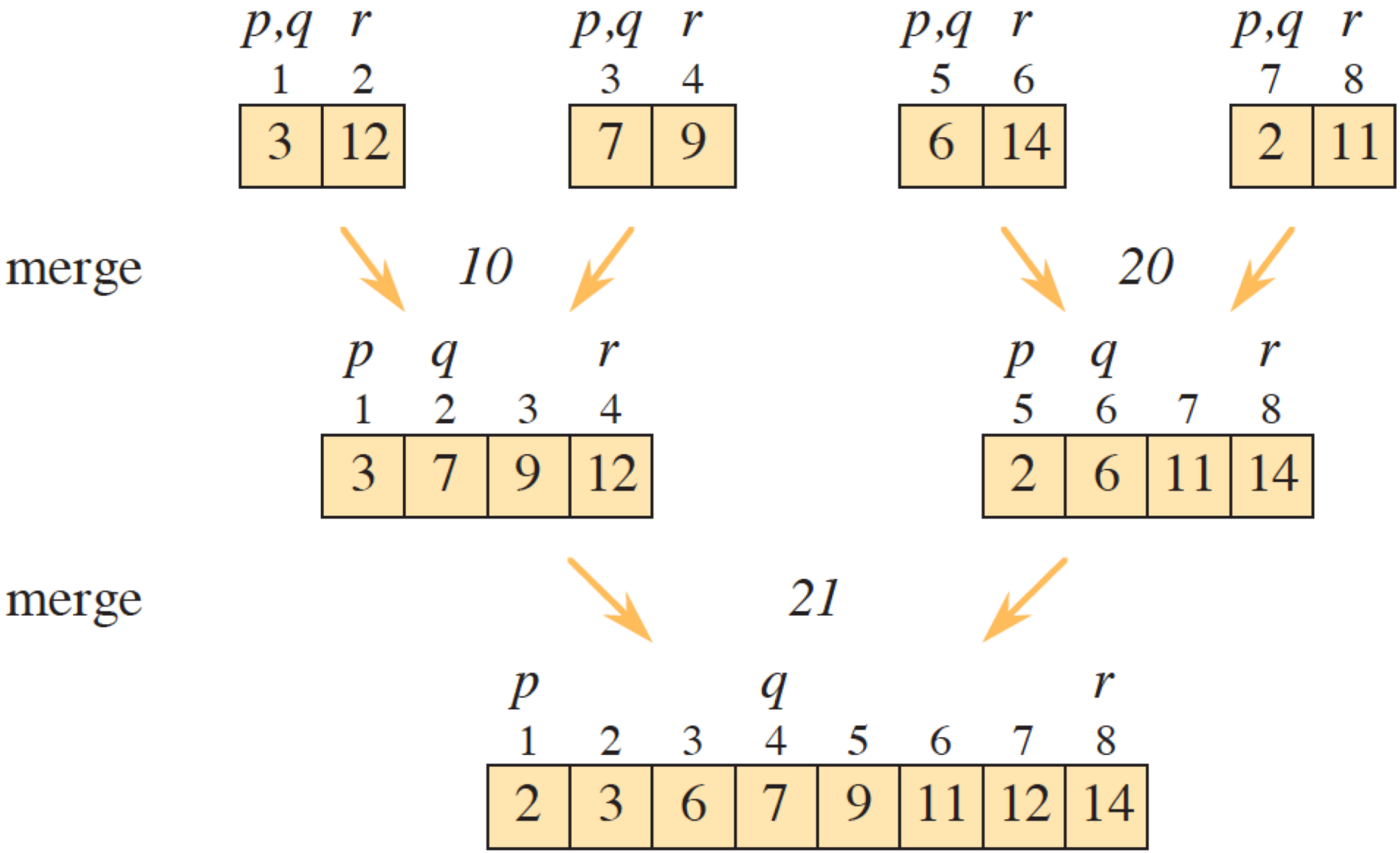
MERGE-SORT example



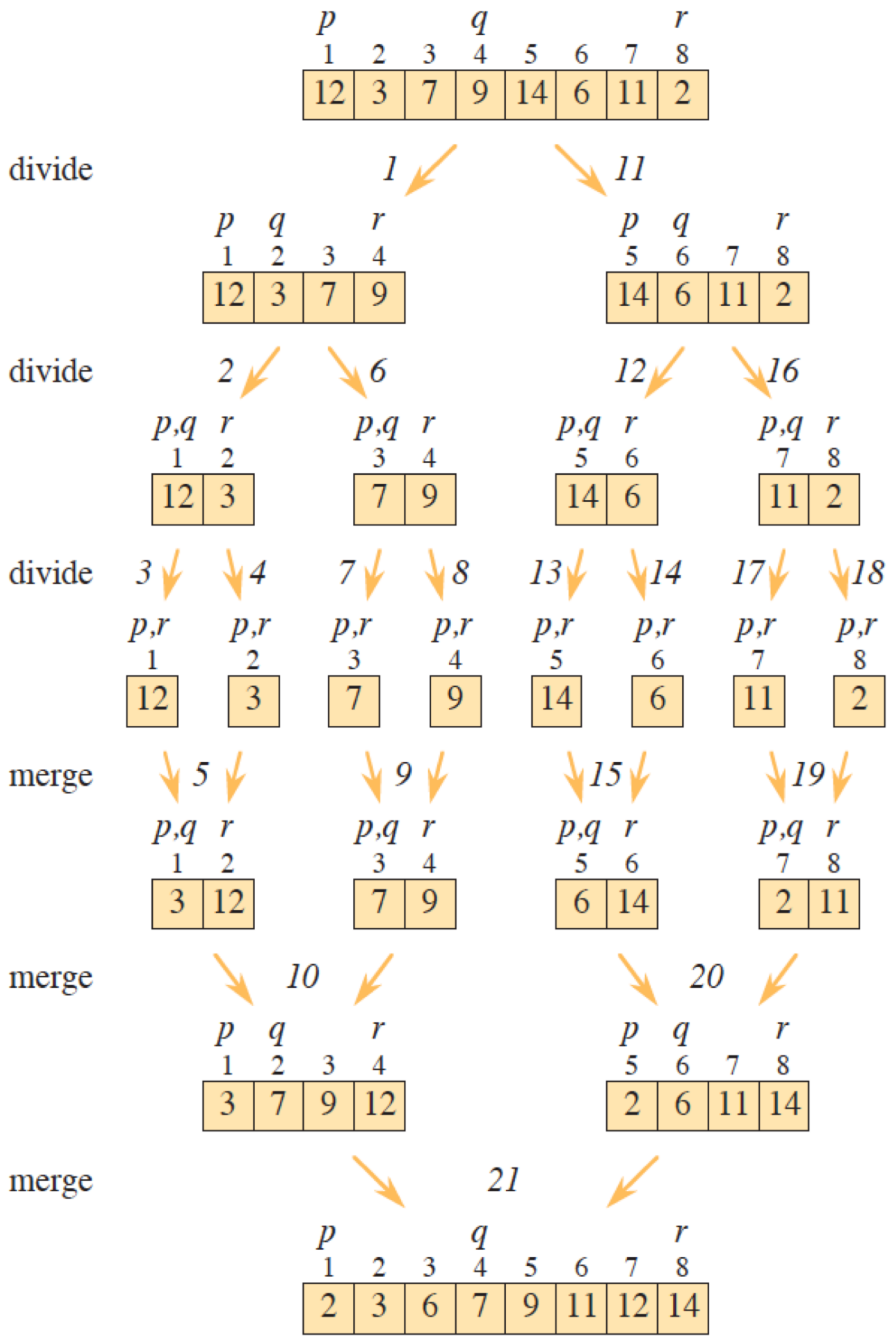
MERGE-SORT example cont'd



MERGE-SORT example cont'd



MERGE-SORT example cont'd



The merge algorithm

- The key operation of the merge sort algorithm occurs in the combine step, which merges two adjacent, sorted subarrays.
- The merge operation is performed by the auxiliary procedure $\text{MERGE}(A, p, q, r)$
- The procedure assumes that the adjacent subarrays A[p:q] and A[q+1:r] were already recursively sorted.
- It merges the two sorted subarrays to form a single sorted subarray that replaces the current subarray A[p:r].

$$p \leq q < r$$

The procedure MERGE

```
MERGE(A, p, q, r)
 1  n_L = q − p + 1          // length of A[p : q]
 2  n_R = r − q              // length of A[q + 1 : r]
 3  let L[0 : n_L − 1] and R[0 : n_R − 1] be new arrays
 4  for i = 0 to n_L − 1     // copy A[p : q] into L[0 : n_L − 1]
 5      L[i] = A[p + i]
 6  for j = 0 to n_R − 1     // copy A[q + 1 : r] into R[0 : n_R − 1]
 7      R[j] = A[q + j + 1]
 8  i = 0                    // i indexes the smallest remaining element in L
 9  j = 0                    // j indexes the smallest remaining element in R
10  k = p                    // k indexes the location in A to fill
```

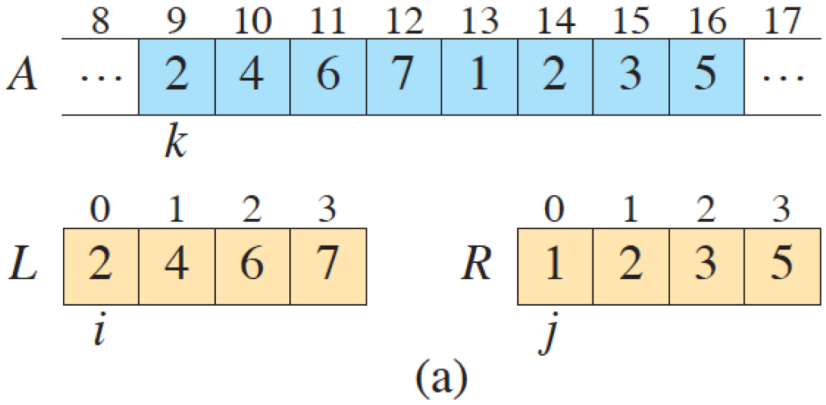
The procedure MERGE cont'd

```
11  // As long as each of the arrays L and R contains an unmerged element,
    //     copy the smallest unmerged element back into A[p : r].
12  while i < n_L and j < n_R
13      if L[i] ≤ R[j]
14          A[k] = L[i]
15          i = i + 1
16      else A[k] = R[j]
17          j = j + 1
18      k = k + 1
```

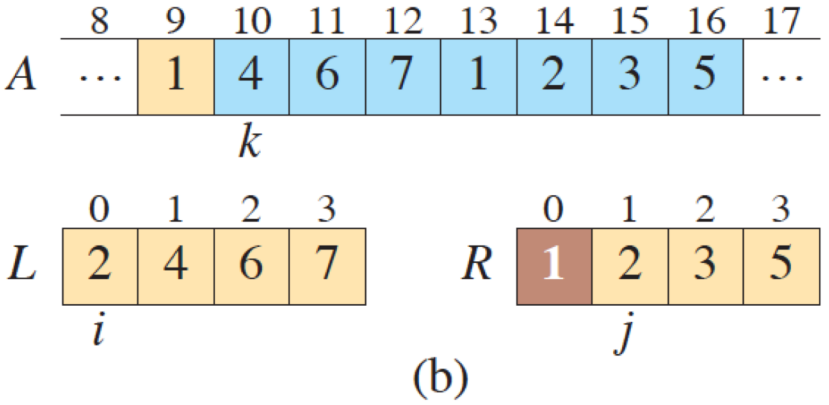
The procedure MERGE cont'd

```
19  // Having gone through one of L and R entirely, copy the
    //      remainder of the other to the end of A[p : r].
20  while i < n_L
21      A[k] = L[i]
22      i = i + 1
23      k = k + 1
24  while j < n_R
25      A[k] = R[j]
26      j = j + 1
27      k = k + 1
```

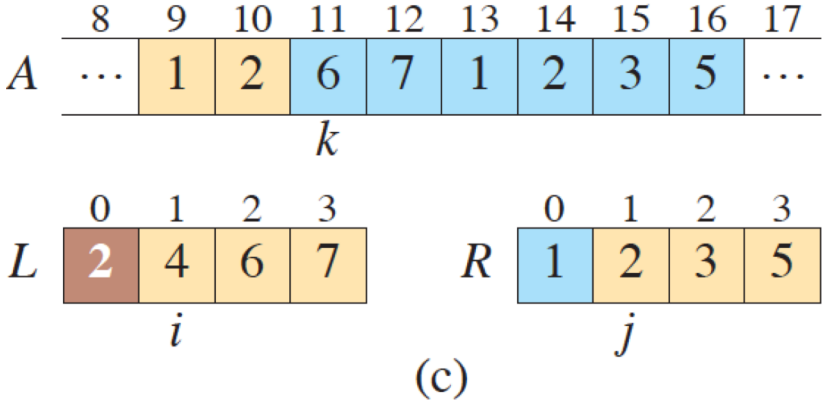
MERGE example



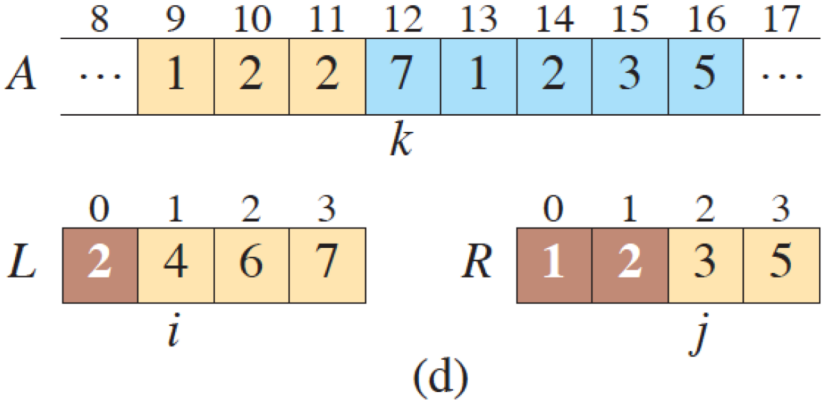
(a)



(b)

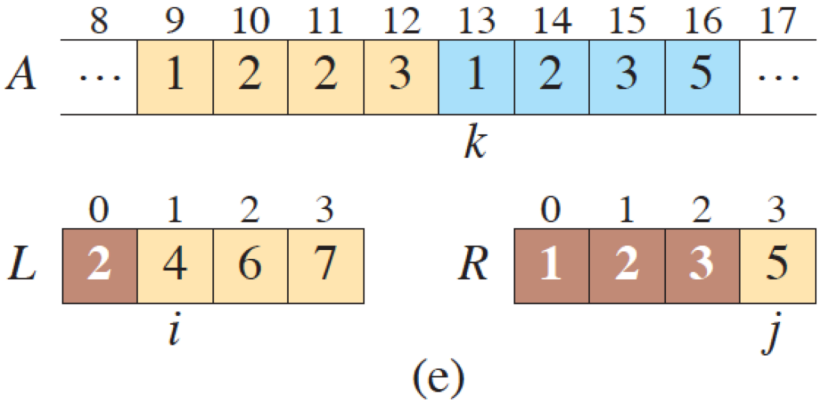


(c)

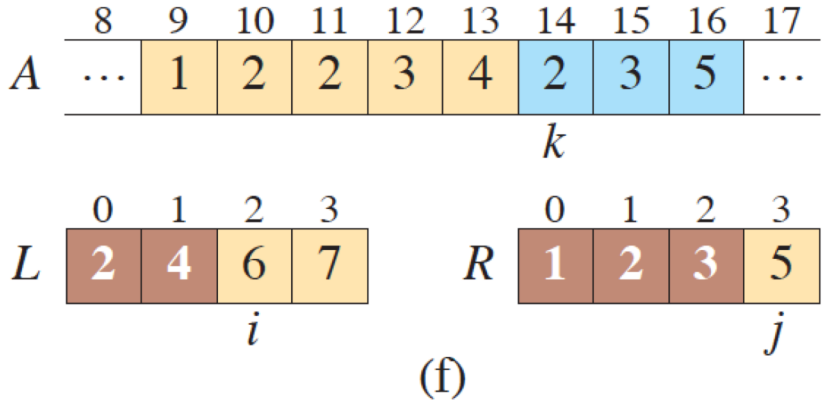


(d)

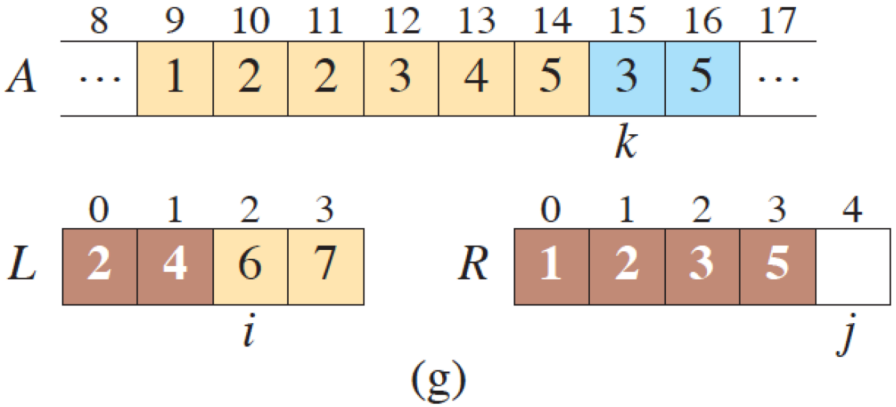
MERGE example cont'd



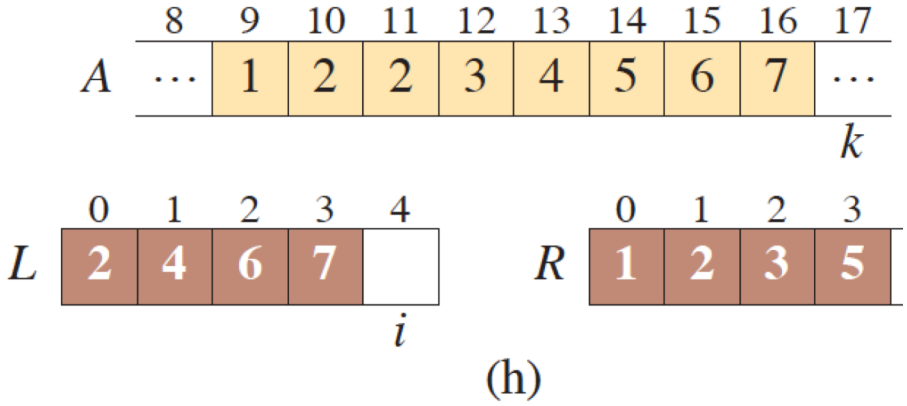
(e)



(f)



(g)



(h)

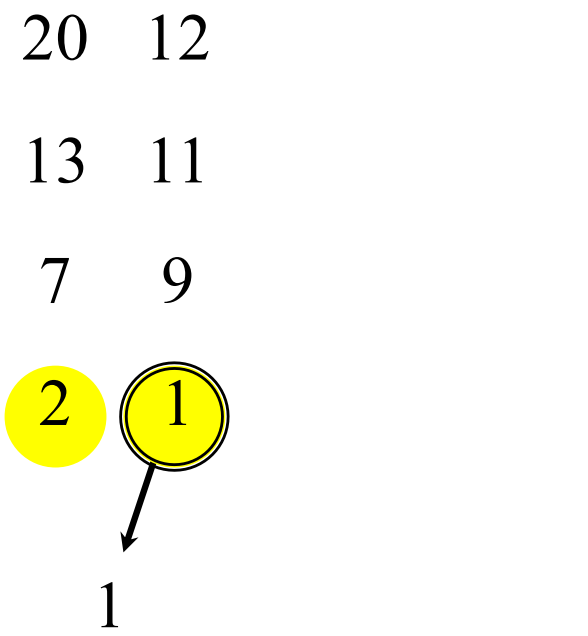
MERGE running time

To see that the MERGE procedure runs in $\Theta(n)$ time, where $n = r - p + 1$,[13] observe that each of lines 1–3 and 8–10 takes constant time, and the **for** loops of lines 4–7 take $\Theta(n_L + n_R) = \Theta(n)$ time.[14] To account for the three **while** loops of lines 12–18, 20–23, and 24–27, observe that each iteration of these loops copies exactly one value from L or R back into A and that every value is copied back into A exactly once. Therefore, these three loops together make a total of n iterations. Since each iteration of each of the three loops takes constant time, the total time spent in these three loops is $\Theta(n)$.

Merging two sorted arrays

20	12
13	11
7	9
2	1

Merging two sorted arrays



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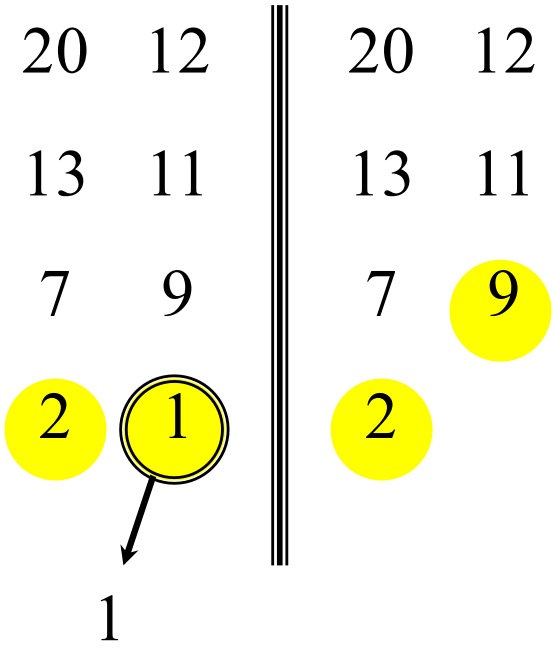

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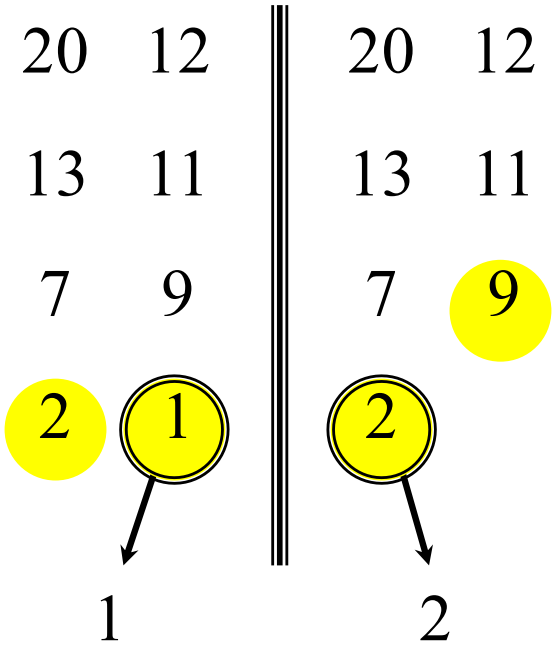

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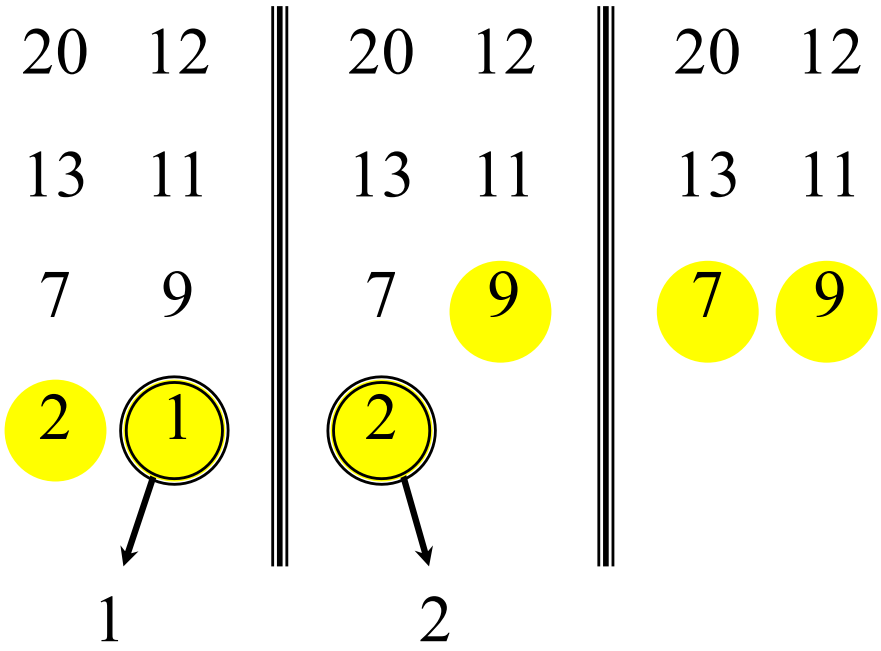

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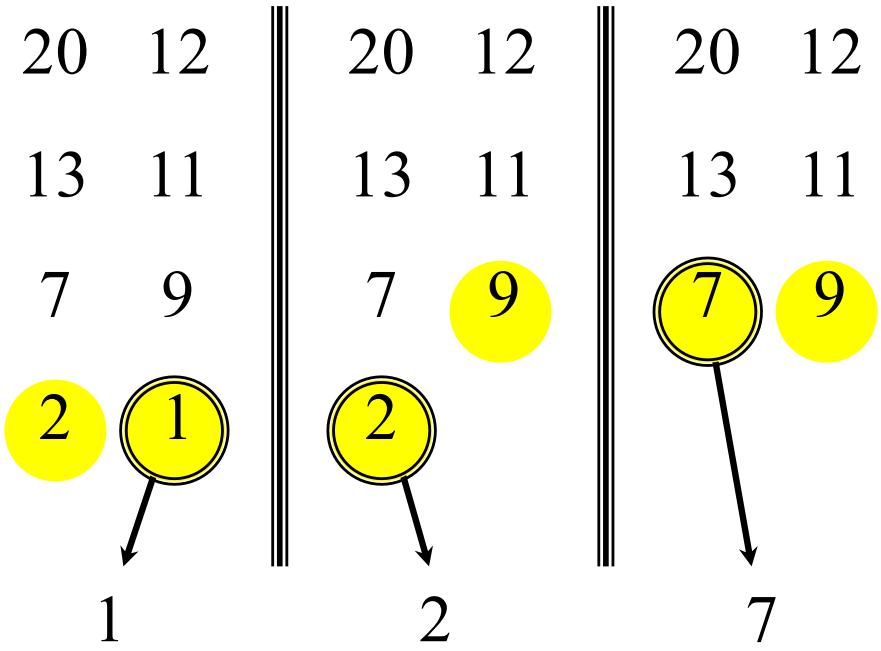

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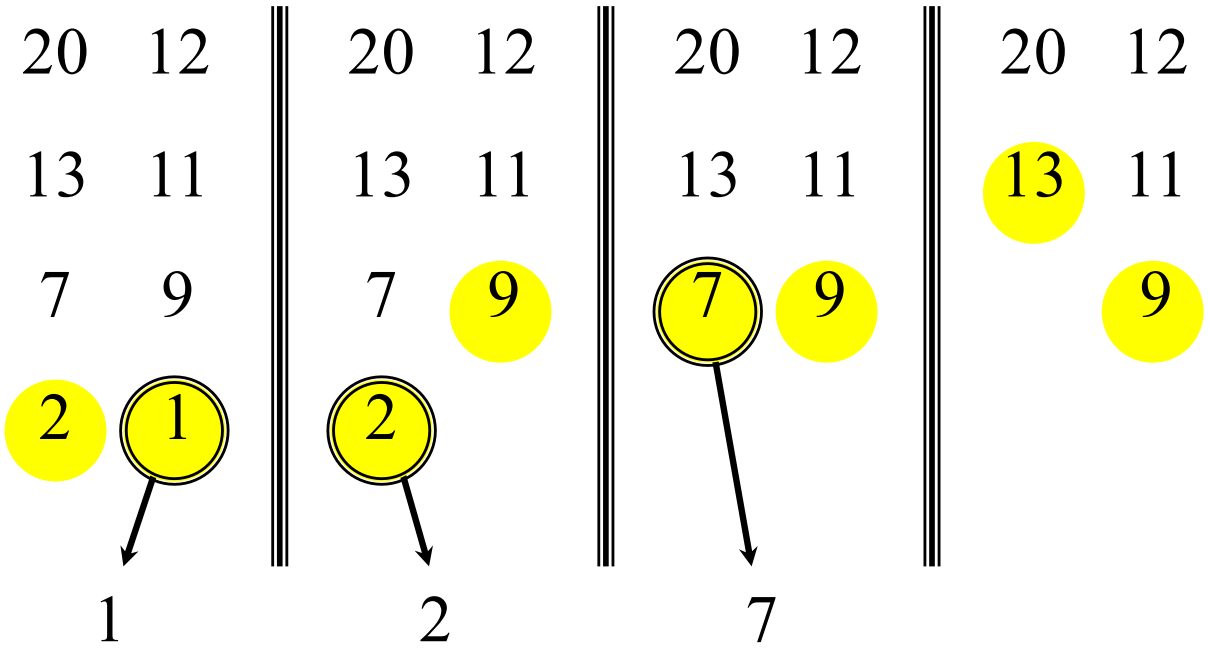

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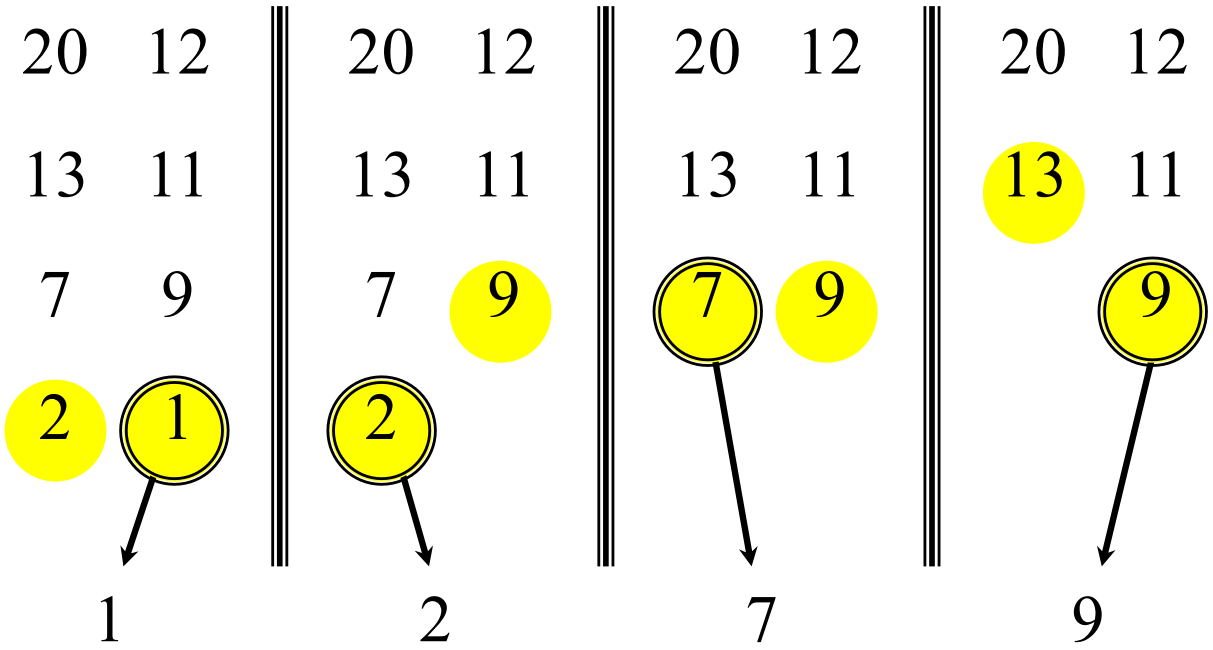

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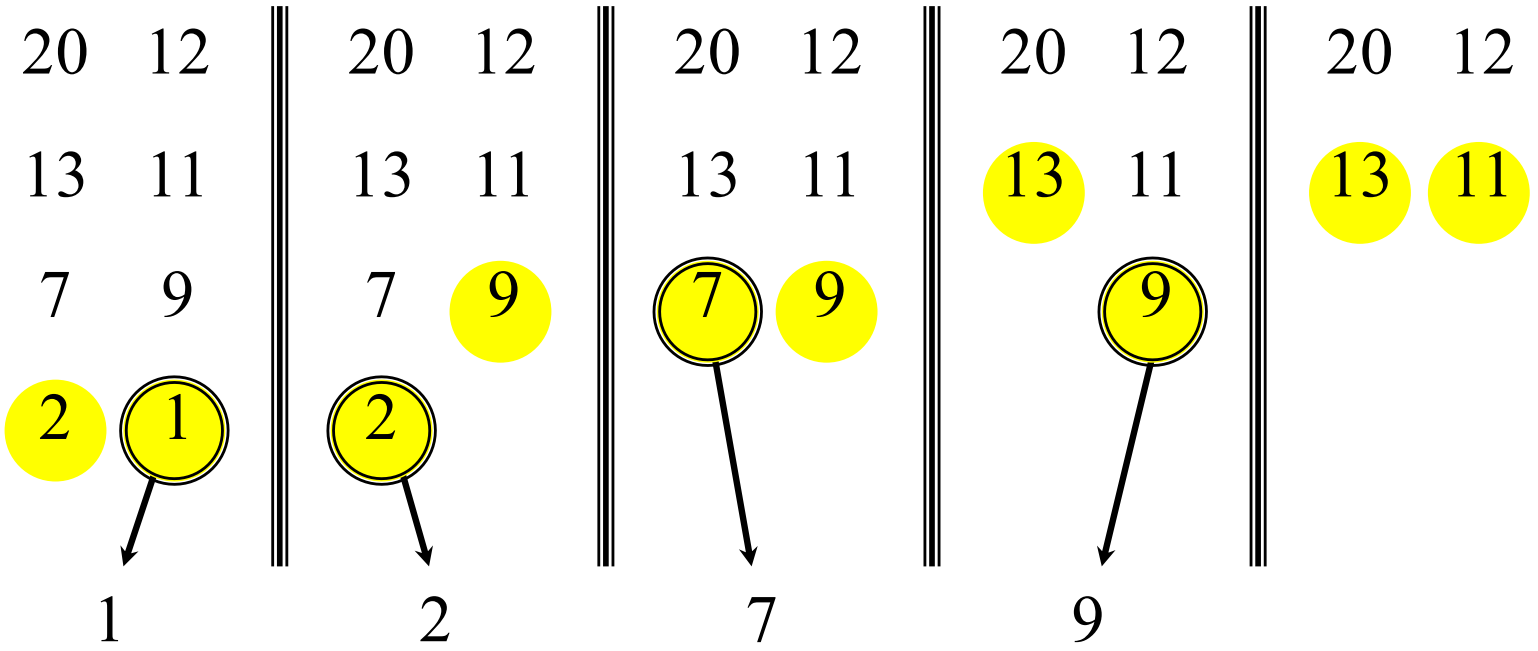

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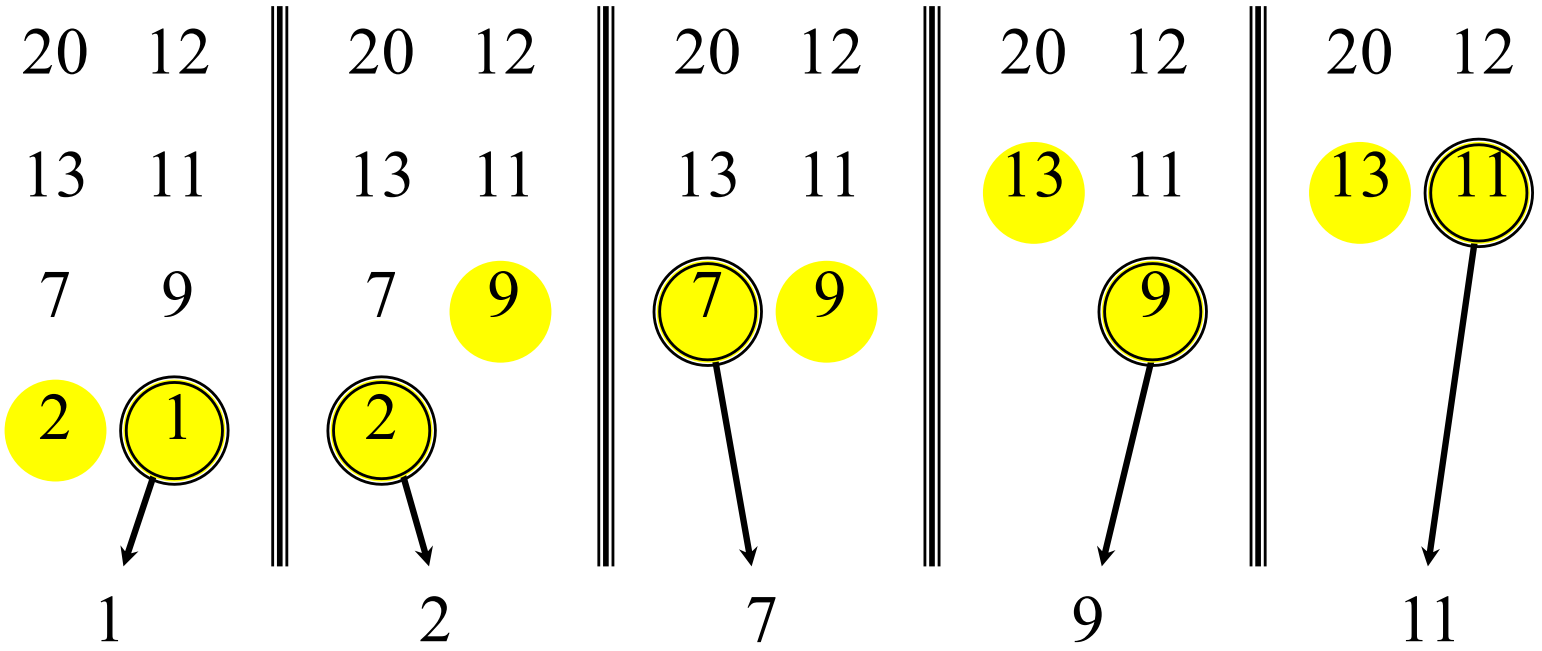

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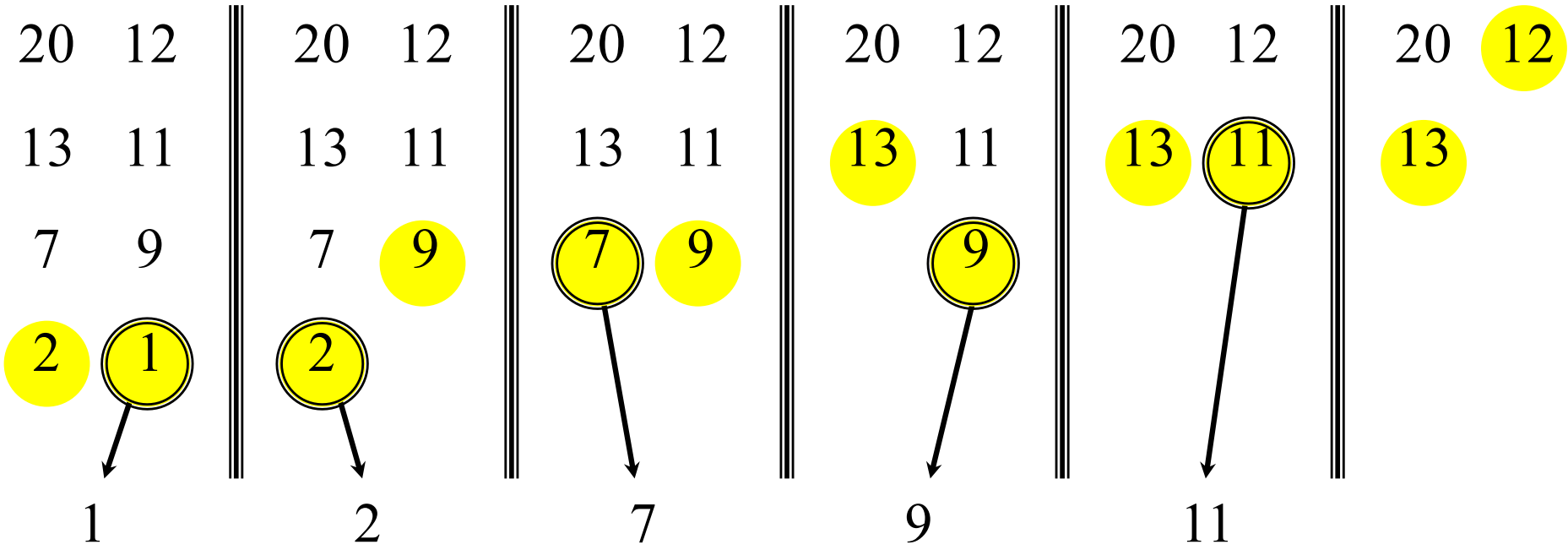
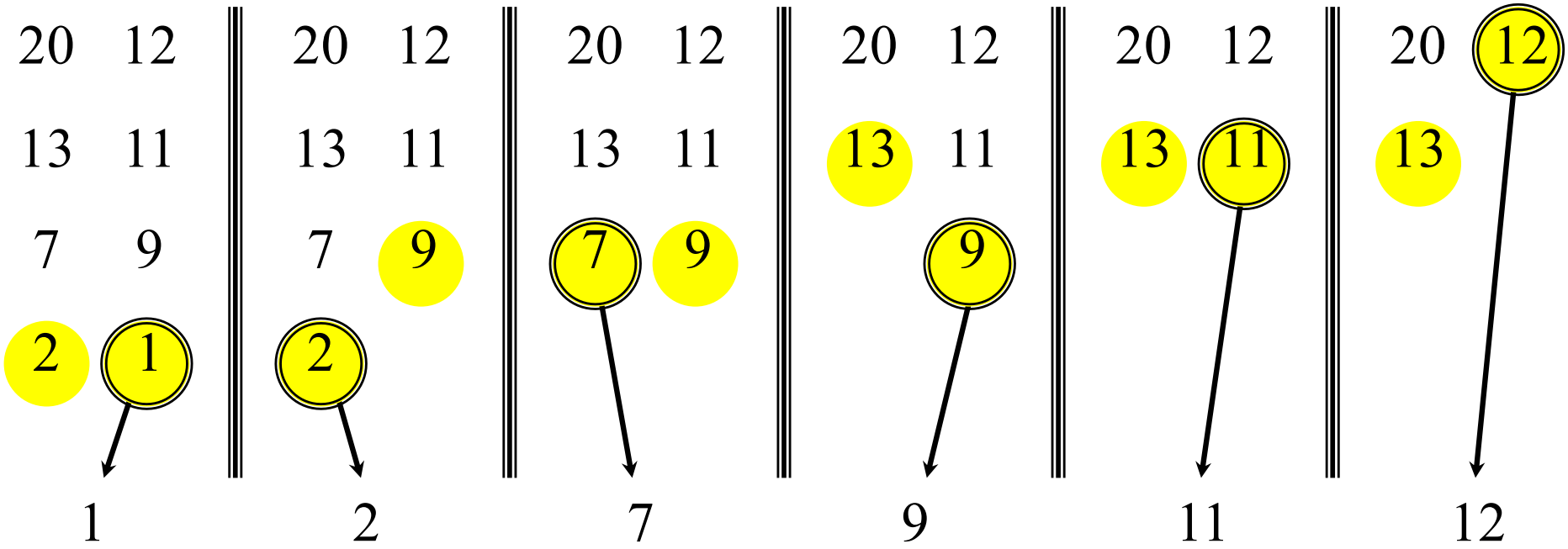


Merging two sorted arrays



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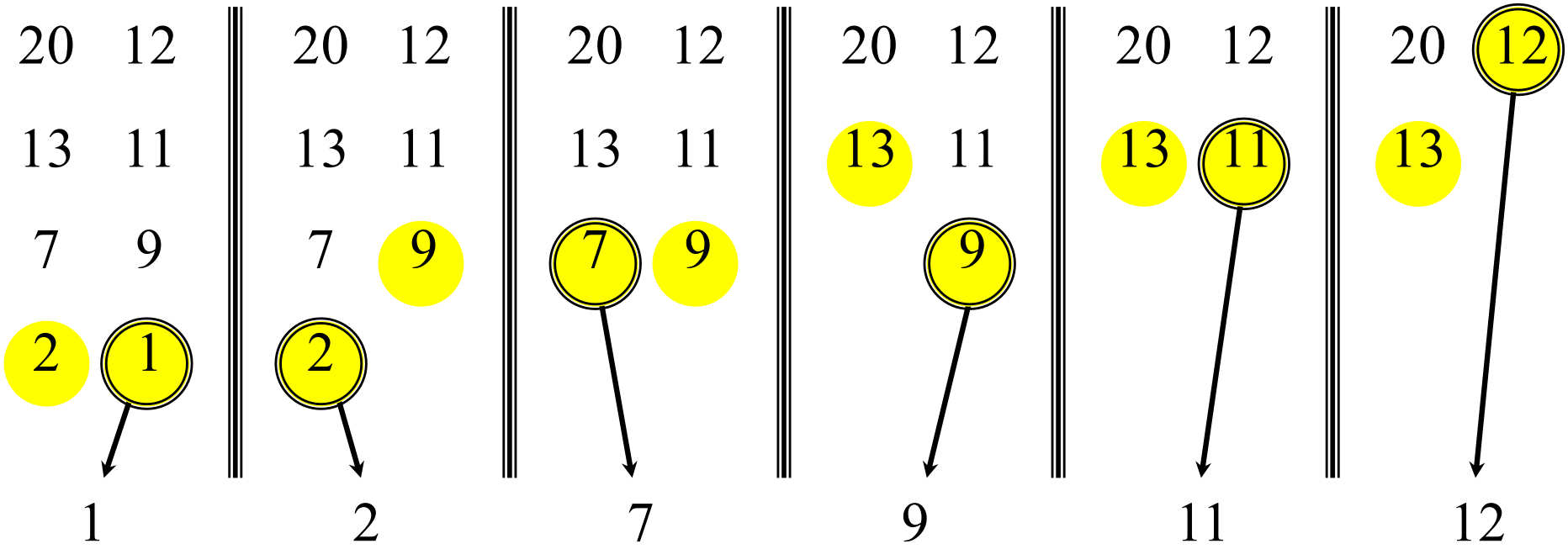


Time = $\Theta(n)$ to merge a total of n elements (linear time).

MERGE-SORT running time

Analysis of merge sort

Here's how to set up the recurrence for $T(n)$, the worst-case running time of merge sort on n numbers.

Divide: The divide step just computes the middle of the subarray, which takes constant time. Thus, $D(n) = \Theta(1)$.

Conquer: Recursively solving two subproblems, each of size $n/2$, contributes $2T(n/2)$ to the running time (ignoring the floors and ceilings, as we discussed).

Combine: Since the MERGE procedure on an n-element subarray takes $\Theta(n)$ time, we have $C(n) = \Theta(n)$.

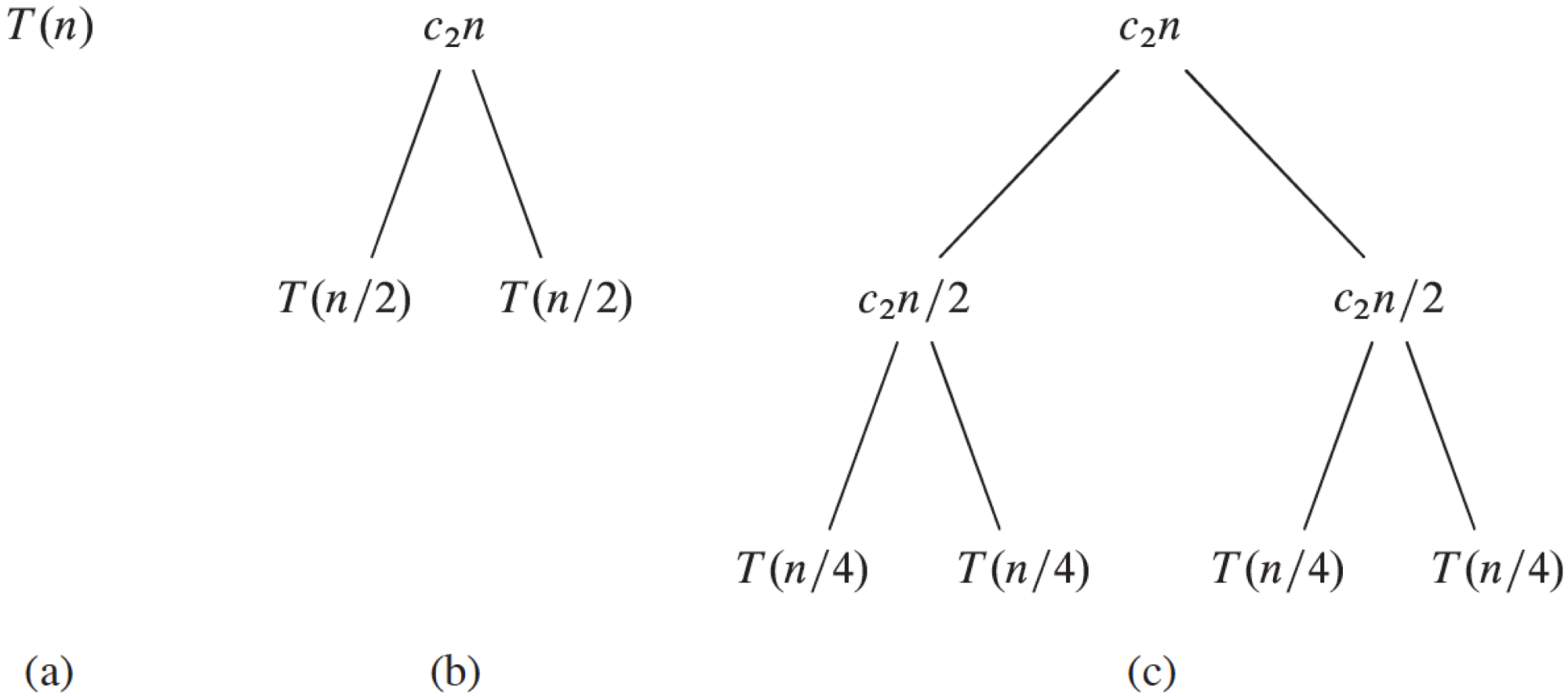
MERGE-SORT running time

$$T(n) = 2T(n/2) + \Theta(n)$$

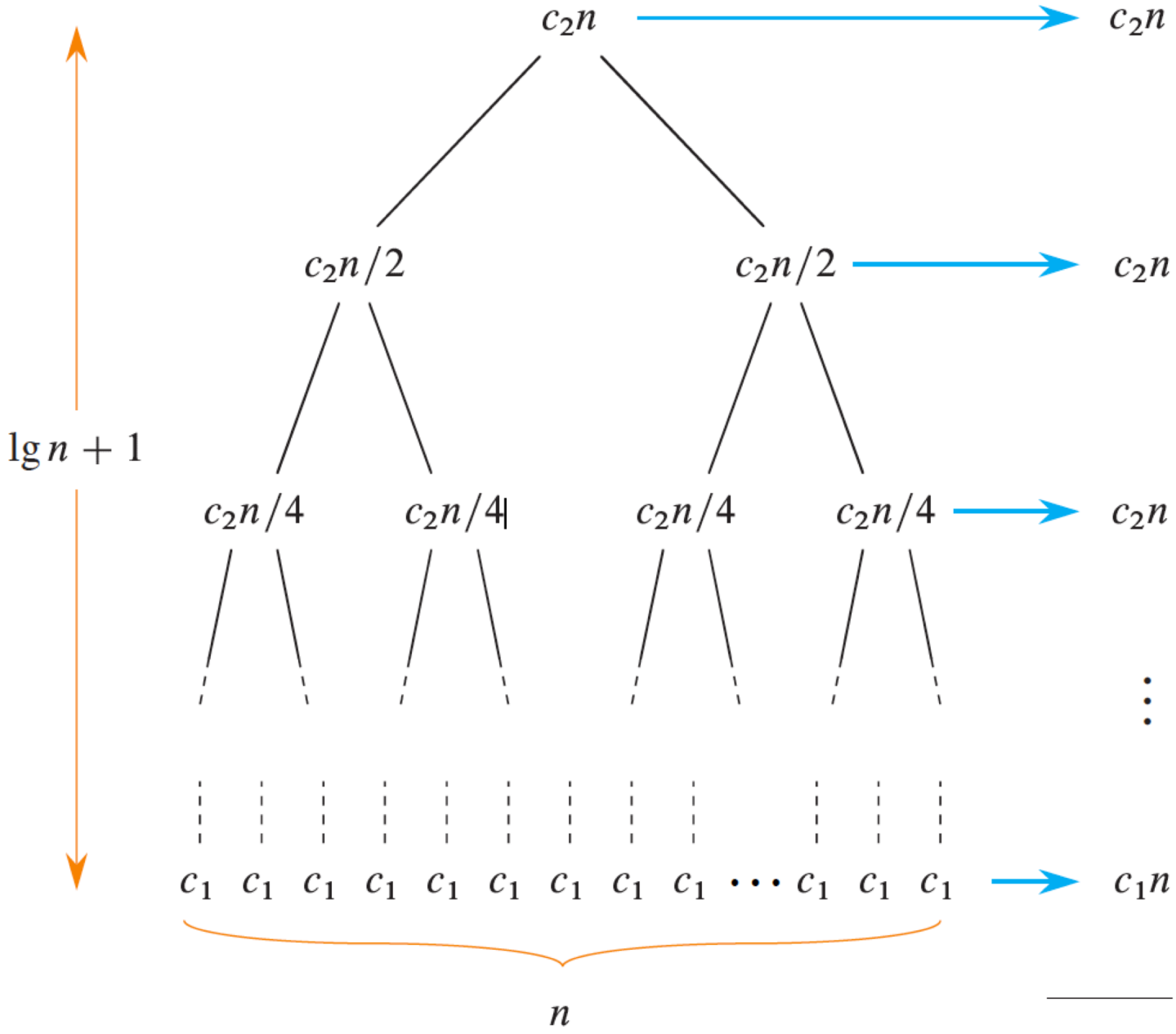
$$T(n) = \begin{cases} c_1 & \text{if } n = 1 , \\ 2T(n/2) + c_2 n & \text{if } n > 1 , \end{cases}$$

MERGE-SORT running time

Recursion tree:



MERGE-SORT running time



MERGE-SORT running time

$$c_2 n \lg n + c_1 n = \Theta(n \lg n).$$

$$T(n) = \Theta(n \lg n)$$